

A flexible and robust network generation pipeline for more detailed bicycle infrastructure representations in MATSim

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1 MOTIVATION

Political and planning debates about cycling are to a large extent debates about infrastructure. Typical measures include converting a street into a bicycle road, upgrading an advisory bicycle lane to a protected one, or closing a gap in the cycling network. A transport model that is meant to inform such decisions has to fulfill two requirements. First, it needs a faithful representation of the existing cycling infrastructure, since the effect of a measure can only be as credible as the baseline it is compared against. Second, the measures themselves should be easy to express in the model, ideally as local edits to network attributes.

On the behavioral side, the type of cycling infrastructure, riding comfort, and gradient are among the strongest drivers of cyclists' route choice (Broach et al., 2012; Hardinghaus & Papantoniou, 2020; Pillat et al., 2023). The bicycle contrib of MATSim (Horni et al., 2016) provides bicycle-specific link speeds, routing, and scoring that react to such attributes (Ziemke et al., 2017). In practice, however, the networks used with the contrib rarely contain the required information. MATSim networks are typically generated from OpenStreetMap (OSM) data (OpenStreetMap contributors, 2026), using either an extension of MATSim's standard OSM readers or a process involving SUMO's netconvert tool (Alvarez Lopez et al., 2018). Both were designed with motorized traffic in mind: they merge adjacent edges without considering cycling attributes, drop footways that are open to cyclists, and elevation at most in simplistic terms. As a consequence, much of the achievable model quality for cycling is already determined during network generation.

This contribution presents a network generation pipeline developed as an update of the bicycle contrib. Every link is directly assigned a category from an established, actively maintained classification of cycling infrastructure and thereby receives an explicit, behaviorally meaningful description. Infrastructure measures then reduce to attribute edits, and scoring can attach behavioral weights per infrastructure category instead of interpreting raw OSM tags.

2 A CYCLING-INFRASTRUCTURE CLASSIFICATION

The core component of the new network generation pipeline is a classifier that maps the roughly 39 relevant OSM tags of a way, from `highway` and the `cycleway` variants to traffic signs, onto one of 27 cycling-infrastructure categories. They range from advisory and exclusive lanes on the carriageway and protected bicycle lanes to bicycle roads, separated cycleways, and foot and cycle paths open to cyclists. Rules that depend on traffic signs are country-specific and pluggable; a profile exists for Germany, a first experimental one for Austria, and a generic profile relies on tag-only logic for other countries. In addition, access rules are handled consistently: one-way streets that are open to cyclists in the reverse direction receive a contraflow link, footways and access-restricted ways are opened for cycling only where OSM explicitly permits it, and the bicycle mode is removed where cycling is prohibited. The category, the relevant raw tags (e.g., surface and smoothness), and the elevation attributes described below are stored as additional link attributes in the standard MATSim network format and can be used by any downstream analysis or scoring.

The category system is adopted from the `radinfra.de/TILDA` project (FixMyCity and OSM Verkehrswende Community, 2026). It has matured over several years, is continuously reviewed by the OSM community, and is actively maintained by its authors. Building on it has two advantages beyond the categories themselves. First, the classification is robust with respect to real-world tagging: the same physical infrastructure is often tagged in several different and sometimes imprecise ways, and the classifier resolves such variants to the correct category as long as the tagging remains unambiguous. It thereby accounts for the fact that OSM data is rarely perfect. Second, the approach is extensible, e.g. infrastructure widths can be added with manageable effort.

3 NETWORK GENERATION

The classification can be applied in two ways. The first path targets existing scenarios whose networks are built with SUMO's netconvert tool, in recent years the common approach for most MATSim scenarios built in Germany, including the more recent version of the Open Berlin scenario. Two commands annotate the finished network in place, so that the classification becomes part of the scenario network itself. Care is needed because netconvert merges consecutive edges without considering cycling tags, so a merged edge can carry the cycling attributes of only one of its constituent ways. The tools therefore resolve all attributes from the original OSM ways, and a second netconvert pass ensures that edges are never merged across category boundaries.

The second path is a standalone pipeline that converts an `.osm.pbf` extract into a complete MATSim network with a single command. The classification runs during the OSM import. A bicycle-aware simplification then merges consecutive links only if they agree on all infrastructure-relevant attributes, whereas the default simplifier merges across infrastructure changes and thereby loses the attached information. Both paths produce the same cycling-related attributes. Other network properties differ; in particular, the detailed freespeeds and capacities are currently only derived in the SUMO path.

Both paths also take elevation into account. Gradients are sampled from high-resolution terrain models, such as the LiDAR-based Sonny DTMs¹, along the actual link geometry; each link receives an end-to-end and a maximum gradient as well as its total elevation gain and loss.

4 APPLICATION TO THE BERLIN SCENARIO

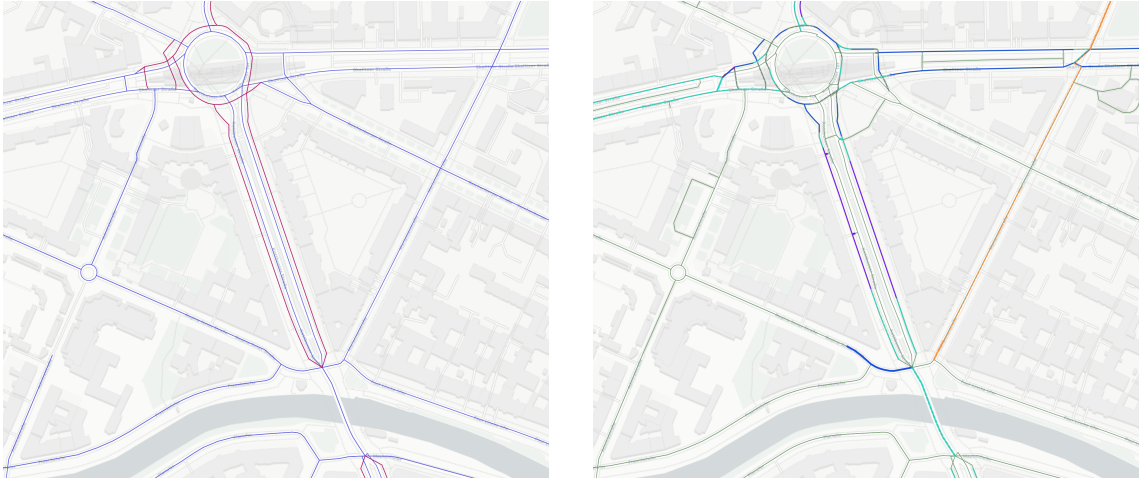
We applied the SUMO pipeline to Berlin, using an OSM extract provided by Geofabrik (2026). Figure 1 compares the representation of cycling infrastructure around Kottbusser Tor in the Open Berlin Scenario version 7.0 network and in the network generated our new pipeline. In version 7.0, some links with dedicated cycling infrastructure can be identified, but without further differentiation. Previously fragmented connections, such as the path in the western part of the map cutout, become continuous and thus usable for cycling. The new network distinguishes the infrastructure categories such as the bicycle road is shown in orange on the right.

The generated network covers Berlin and a 15 km buffer in full detail, embedded in a coarser network for the surrounding region. For a fair comparison, we clip both networks to the Berlin city limits and remove links exclusive to public transport: within the city, the number of links grows from about 84,000 in version 7.0 to 173,000, and the total length from 11,800 km to 15,700 km.

The road network itself is essentially unchanged (10,940 versus 10,890 km from `residential` up to `motorway`), but is represented by more and shorter links (78,000 versus 108,000). The growth that matters for cycling lies in the minor ways: service roads, footways, paths, tracks, and separate cycleways grow from 5,600 links (870 km) to 65,400 links (4,770 km). These links close gaps and make the network usable for cycling, although most of them do not carry dedicated cycling infrastructure themselves.

Within the city, about 48,000 links totalling 3,700 km carry a cycling-infrastructure category (28 % of all links, 24 % of the length), among them some 530 km of bicycle lanes on the carriageway, 1,040 km of cycleways, and 100 km of bicycle roads; only 0.9 % of all links are marked as needing additional clarification by the process.

¹<https://sonny.4lima.de>



(a) Berlin scenario 7.0

(b) New pipeline, classified infrastructure

Figure 1: Cycling infrastructure around Kottbusser Tor in Berlin-Kreuzberg. (a) Open Berlin Scenario version 7.0 network; links with dedicated cycling infrastructure are shown in red. (b) Network generated by our new pipeline, with links colored by cycling-infrastructure category; the bicycle road, for example, is shown in orange. Map data © OpenStreetMap contributors.

87 The same area can be inspected in the interactive map of the radinfra.de project (FixMyCity
88 and OSM Verkehrswende Community, 2026), which we used for visual plausibility checks of the
89 classification.²

90 5 OUTLOOK AND CONCLUSIONS

91 A second development line adapts the bicycle contrib so that it can make use of the new attributes.
92 Scoring and speed parameters are moved behind a pluggable interface, and the infrastructure factor
93 can be based directly on the 27 categories; a configuration switch keeps existing setups working.
94 Detailed and trustworthy network attributes are a prerequisite for answering questions about cy-
95 cling infrastructure with MATSim. The presented pipeline provide these attributes in a repro-
96 ducible and tested way for two common network generation workflows, and they turn infrastructure
97 measures into simple attribute edits. The code is currently developed on branches of matsim-libs;
98 integration into the bicycle contrib via pull request is planned. Feedback from the user community
99 is welcome. We intend to present this work in person.

ACKNOWLEDGMENT

This work is part of the Collaborative Research Centre (CRC) AgiMo, which investigates data-driven approaches to responsible mobility planning and relies on high-quality geospatial data as a foundation for mobility modelling. The work is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) — TRR: 408/1 2025 — Project number: 531327426.

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²Permalink: <https://tilda-geo.de/regionen/radinfra?map=17/52.4976/13.4192&config=1p2va4k.7h3d.9fm70g&v=2>

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